

PaliGemma · arXiv 2407.07726 · Sep 16

Empiricist

seat 3 · 11+3

Res ladder · LoRA freeze matrix · GPU budget

Paper club · VLA Architectures · dark academic

Hypotheses · Empiricist

Cluster-feasible pressure tests of transfer claims

H1

448pt + LoRA $r=16$ closes $\geq 70\%$ of
224→448 OCR/VQA gain at $< 30\%$
of full-FT cost (VRAM/wall-clock)

H2

Native Stage2-res ckpt beats
cross-res FT under LoRA
(replicates Fig.10 qualitative)

H3

K-frame concat (4-8 @224) helps
short-horizon BEHAVIOR-style VQA
vs single frame (fixed token budget)

Success = qualitative Table1 deltas + honest VRAM/steps — not paper ImageNet parity

Log: resolution, freeze flags, LoRA r , effective batch, steps, peak VRAM, dupe-filter

Resolution ladder · P0 E1

OCR-ish tasks: DocVQA / TextVQA — where pixels + tokens both matter



Experiment: LoRA r=16 on Gemma; connector FT; freeze ViT first pass; same step budget @224 vs 448

H1 target: close $\geq 70\%$ of 224→448 gain at $< 30\%$ full-FT cost

to read *all* text on the image in left-to-right, top-to-bottom order. For detection and segmentation tasks, we can request the model to detect or segment *all* objects for which annotation is provided. Hence, we also increase the text sequence length to $N_{\text{txt}} = 512$ tokens.

While PaLI has always had this resolution increasing stage, and for image classification the importance of resolution is long known [55, 109], several recent works [81, 114, 121] have raised the importance of resolution in VLMs too. We add to this body of knowledge by providing several ablation studies regarding Stage2 in Section 5.7.

3.2.4. Stage3: Transfer

The result of Stages 1 and 2 is a family of three PaliGemma checkpoints, at 224px, 448px, and 896px resolution, which are pre-equipped with broad visual knowledge. However, these checkpoints are not “user (or benchmark) friendly” as their pretraining has focused solely on density of learning signal, as opposed to usable interface.

These base models need to be transferred to serve their intended final purpose. That could take the form of fine-tuning on a specific, specialized task, such as COCO Captions, Remote Sensing VQA, Video Captioning, or InfographicQA. Adapt to new inputs such as multiple images (NLVR2) or bounding boxes draw in the image (WidgetCap). Or it could take the form of instruction [70] or even chat [46] tuning.

To show the effectiveness of the base models, we transfer them to a wide range of individual academic benchmarks, using a simple unified transfer recipe with few hyper-parameters. And to showcase the versatility beyond academic tasks, we also provide a “mix” transfer checkpoint, which transfers to a subset of these tasks at the same time, along with detailed captioning and long question-answering data. While this is not instruction tuning, it is a step in that direction.

We also transfer PaliGemma to tasks which take multiple images as input. NLVR2 is one such task, which asks one question about two images, and requires looking at both to give the correct answer. Other such tasks are standard short-video

understanding tasks subsampled to 16 frames. In all these cases, we follow PaLI-3 and encode each image separately, then concatenate the image tokens without any special separator or embedding tokens. Thus, 16 frames at 224px resolution result in $N_{\text{img}} = 4096$ image tokens, the same amount as a single image at 896px resolution.

For all transfers, we perform fine-tuning of all the model parameters. The hyper-parameters we modify per-task are the following, in decreasing order of importance:

- Resolution (*i.e.* checkpoint): **224**, 448, 896.
- Epochs: **1**, **3**, **10**, 30, 100.
- Learning-rate: 3e-5, **1e-5**, 3e-6.
- Label-smoothing: **0.0**, 0.1, 0.3.
- Dropout in the LLM: **0.0**, 0.1, 0.3.
- Weight decay: **0.0** or $0.1 \times \text{learning-rate}$.
- Freeze ViT: **false**, true.
- Beam-search may benefit captioning.

The above are typical values we suggest exploring, with the recommended initial attempt value in bold. We provide the best setting for each individual task in Appendix J. We study the sensitivity to transfer hyper-parameters in Section 6.2, and the “transferability” in general in Section 6, showing that good results can be achieved with the aforementioned initial attempt values.

3.2.5. Pretraining task mixture

Just like for previous PaLI models, the pretraining (Stage1 and Stage2) is designed to result in a model that transfers well, not necessarily a model that is usable out of the box (“0 shot”). The intuition here is that we want a mix of tasks which force the model to acquire a broad range of “skills”. We prefix each task with its unique prefix to avoid conflicting learning signals across skills [14]. At transfer time (Stage3), the model then merely needs to recognize which skill is useful for the task, and rewire itself to use that while following the output syntax and vocabulary of the task. In our experience, these can all be done relatively quickly and based on few examples (Section 6.3). We do not use any of our transfer datasets during pretraining, and furthermore remove all near-duplicates of their images from

LoRA freeze matrix · P0 E2

TT / TF / LoRA-LLM / LoRA+connector — four short runs @224

TT

Train All

Full FT
ViT + LLM +
connector

Highest VRAM
Paper default spirit

TF

Freeze ViT

LLM + connector
ViT locked

ScienceQA-like
≤2pt VQA cost?

LoRA-LLM

PEFT LLM

$r=16$ attn
ViT frozen

~24GB target
H1 workhorse

LoRA+

LLM+conn

LoRA + unfreeze
projector

Best cost/quality
bet for transfer

Paper §5.4: TT $\approx$ TF on avg transfer; unfreeze ViT helps spatial; freeze LM hurts a lot

P0 E3: default HP ($\text{lr}=1\text{e-}5$, $\text{bs_eff}=256$ via accum, no LS/DO) vs tiny sweep regret
Outlines: /workspace/paligemma-quiet/env_outline/

GPU budget honesty

Paper is TPU-only — these are derived Empiricist estimates (~2.92B params)

LoRA r=16 @224

~12-20 GB

Good 24GB card

LoRA @448

~20-35 GB

40-48GB comfortable

Full FT @224

~40-60+ GB

Often 80GB or ZeRO

Full FT 448-896

80GB-multi-GPU

Match paper spirit costly

16×224 video

~40-80 GB

Like single 896 tokens

Headline for room: LoRA@224 ~24GB · full FT 448-896 needs 80GB+

P0 experiment plan

E1

Resolution ladder

224 / 448 / (896)
on OCR-ish subset
LoRA $r=16$

E2

Freeze matrix

TT / TF /
LoRA-LLM /
LoRA+connector

E3

Default-HP regret

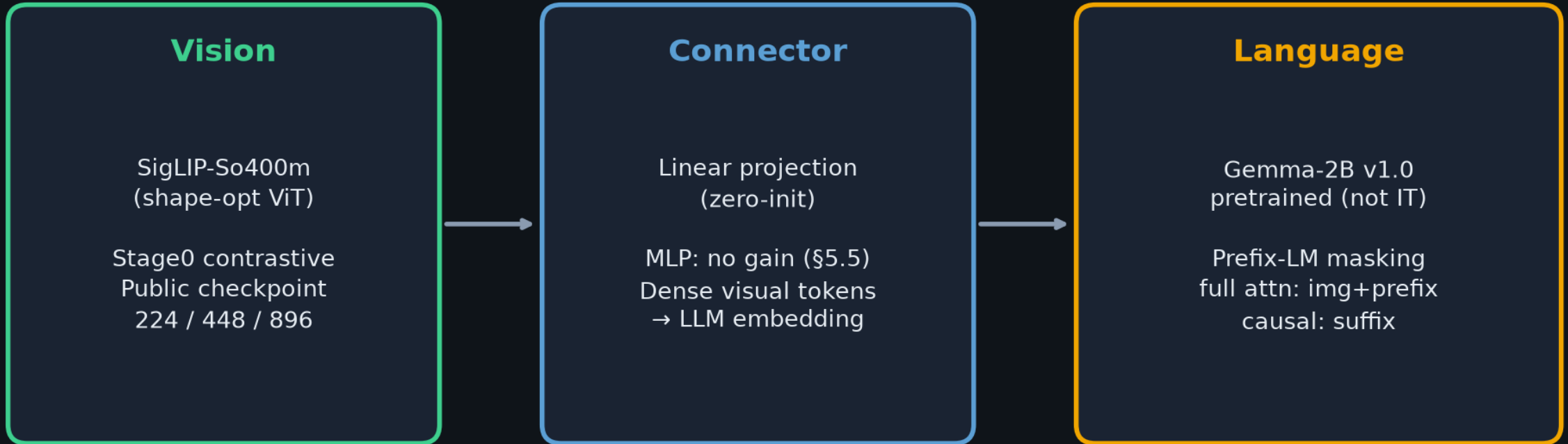
$lr=1e-5$ recipe vs
tiny sweep on
VQAv2/GQA-small

P1 next: LoRA rank · cross-res transfer · few-shot curves

P2 stretch: RefCOCO · multi-frame · encoder-free smoke
Week-slice: 1-2 GPU-days on 5-20% Doc/TextVQA + tiny VQAv2 minival

Architecture · open base ~3B VLM

SigLIP-So400m + linear connector + Gemma-2B · Prefix-LM



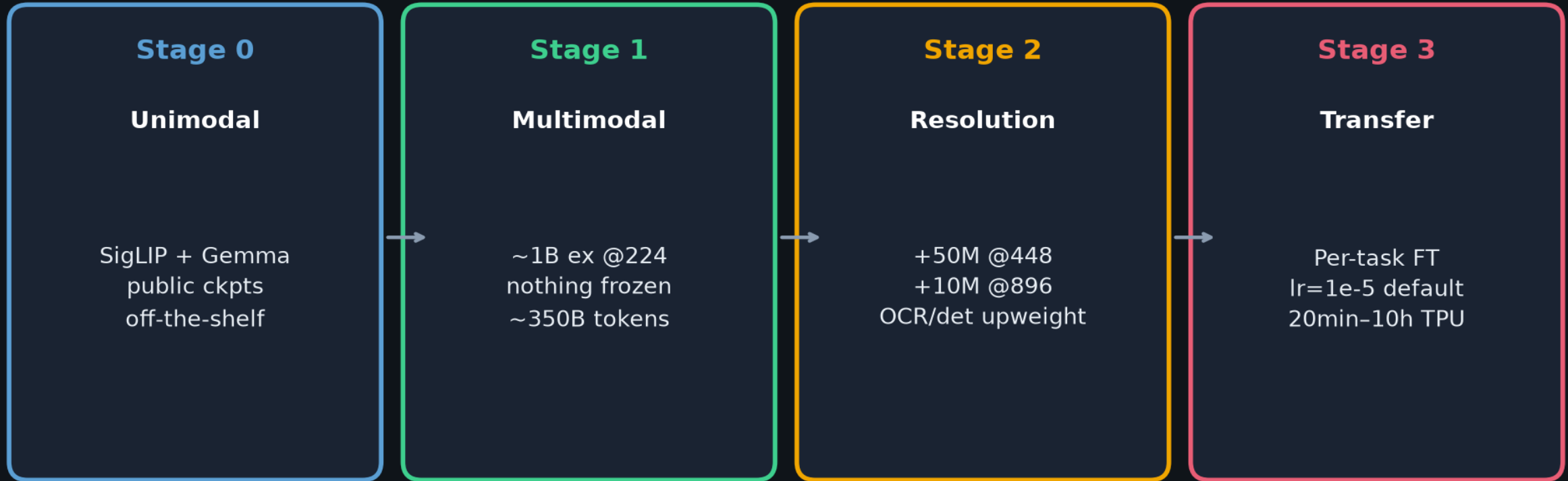
Token layout: [image...] BOS prefix... SEP(\n) suffix... EOS PAD...

loc tokens: 1024 bins · seg tokens: 128 VQVAE codes · skills via task prefixes (no GPT-4 IT)

Perception / grounding backbone — NOT a classical planner or policy

Training stages 0 → 3

Unimodal priors → multimodal skills → resolution → per-task transfer



Key Stage1 choice: do NOT freeze ViT (slow LR warmup) — freezing LM hurts a lot

Prefer released paligemma-pt-{224,448} as substrate for π_0 / OpenPI / BEHAVIOR forks

Empiricist close

H1 LoRA@448 · freeze matrix · 24GB vs 80GB+ honesty